



Ahmedabad
University

CSE623 : Machine Learning

**Machine Learning-based Estimation of Muscle
Cross-Sectional Area (ACSA) from Segmented
Ultrasound Images of Lower Limb Muscles for
Sports Performance Analysis**

Course Instructor: Dr. Mehul Raval
Group Name: NewEveWithAdam
Group Members: Aashaka Ashara
Bansari Jani
Manasvi Gondalia
Aaryan Sheth

Abstract

Proper estimation of the Anatomical Cross-Sectional Area (ACSA) using ultrasound images is significant in determining the state of the muscles and rehabilitation monitoring. Manual analysis is a time-consuming process, which is subject to variation, inspiring automated methods of analysis. Ultrasound images with their masks of segmentation are treated as matrices in this work, $I \in R^{m \times n}$ and $M \in \{0, 1\}^{m \times n}$. We also discuss preprocessing, extraction of Region Of Interest(ROI), and determining geometric and statistical properties to form feature vectors that are consistent in nature and can be used in regression modelling and automated ACSA estimation.

Contents

1	What We Have Done	4
2	What We Aspire to Do	5

1 What We Have Done

- Given the ultrasound image of the Rectus Femoris (RF) and Vastus Lateralis (VL) muscles along with the associated segmentation mask in the form of matrices, where the image is denoted as $I \in R^{m \times n}$ and the mask is denoted as $M \in \{0, 1\}^{m \times n}$.
- Extract the region of interest (ROI) of the muscle by performing element-wise multiplication $R = I \odot M$, which retains only the intensity values within the segmented muscle region.
- Compute geometric and shape properties of the segmentation masks including area, perimeter, compactness, and axis-based measures that describe the morphology of the muscle cross-section.
- Calculate statistical measures from the ultrasound image within the segmented muscle region such as mean intensity, variance, and other texture measures that reflect tissue characteristics.
- Normalize all extracted features and organize them into systematic feature vectors for each sample, stacking them into a feature matrix corresponding to the ground-truth ACSA values.
- Build a feature extraction pipeline to prepare the dataset for further machine learning analysis, enabling subsequent steps such as preprocessing, normalization, regression modeling, performance evaluation, and feature interpretability.

2 What We Aspire to Do

- Code to load the ultrasound images and the segmentation masks into the code, and store the images and masks as matrices.
- Implement image processing functions like noise removal and image intensity normalization to improve image quality before feature extraction.
- Do region-of-interest extraction by multiplying the ultrasound image matrix and the segmentation mask element-wise.
- Introduce a feature extraction module, which will calculate geometric features of the masks, such as area, perimeter, compactness, and the major/minor axis lengths.
- Take statistics intensity measures including mean and variance of the ultrasound image within the area of segmented muscle.
- Take all the features that were extracted and organize them into a formatted dataset and match them with the ground-truth ACSA values.
- Create a Python processing pipeline to prepare the dataset in an organization to further do regression modeling and evaluation.

References

- [1] N. J. Cronin, C. P. Carty, R. S. Barrett, and G. Lichtwark, “Automatic tracking of medial gastrocnemius fascicle length during human locomotion,” *Journal of Applied Physiology*, vol. 111, no. 5, pp. 1491–1496, 2011.
- [2] “Skeletal muscle ultrasound radiomics and machine learning for the earlier detection of type 2 diabetes mellitus,” *Journal of Medical Ultrasound*. [Online]. Available: https://journals.lww.com/jmut/fulltext/2025/04000/skeletal_muscle_ultrasound_radiomics_and_machine.5.aspx